FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS — GRADE 10 | SUB-STRAND 3.1: STATISTICS I	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation assessment. You have spent eight lessons investigating the Maize Price Mystery at Wakulima Market. Now it is your turn to answer the driving question completely and convincingly, using everything you have learned about statistics.

YOUR TASK: Write a full, well-organised explanation that answers — How can we use statistics to tell an honest and useful story about data from our own Kenyan community, and help people make better decisions?

Your explanation must:

- Be written in clear, organised paragraphs (not bullet points alone)
- Reference the Wakulima Market maize flour price data as your main evidence
- Include correctly calculated or described measures of central tendency (mean, median, mode) AND measures of spread (range, and any other relevant measure)
- Include at least ONE correctly drawn or described data display (frequency table, histogram, or frequency polygon)
- Explain how BOTH the family AND the neighbour can be telling the truth at the same time
- Show how an honest statistician chooses the right summary to avoid misleading people
- Connect your statistical reasoning to a real decision a Kenyan family, trader, or government official might make

Read each section prompt carefully. Write your response in the space provided. Use your data tables, graphs, and notes from class. You may use a calculator.

SCORING: Your work will be assessed using the rubric at the end of this document across five criteria. Aim for "Excellent" in every row!

Section 1 — What Is Statistics and Why Does It Matter Here?

In your own words, explain what statistics is and what statisticians do. Then explain why statistics is the right tool	Statistics is the branch of mathematics that involves collecting, organising, displaying, analysing, and interpreting numerical data so that we can draw reliable conclusions and make informed decisions. A statistician does not just report raw numbers — they transform chaotic lists into summaries and visuals that reveal patterns hidden in the data.

for settling the argument between the Nairobi family and their neighbour about maize flour prices at Wakulima Market. Why can't we just 'look at the numbers' without organising them first?	In the Wakulima Market situation, the family and the neighbour are both reporting their honest personal experiences, but personal experience is a very small, unrepresentative sample. The family may have happened to shop on days when prices were at their highest and lowest extremes, while the neighbour may have shopped on 'average' days. Neither experience alone tells the full story of all 40 recorded prices. This is exactly why we need statistics. When we look at the raw list of 40 prices, our eyes cannot detect patterns — numbers like 98, 115, 102, 87, 119 jump around with no obvious order. Statistics gives us tools to organise this chaos: we can sort the data, group it into a frequency table, calculate a single representative value, and measure how spread out the prices really are. Only then can we give a fair, evidence-based answer to the argument — one that neither the family nor the neighbour could reach on their own.

Section 2 — Organising and Displaying the Data

Describe how you organised the 40 maize flour prices from Wakulima Market. Explain the steps you took to build a frequency table (including your choice of class intervals). Then describe the histogram or frequency polygon you drew from that table. What patterns or features became visible in your graph that were completely invisible in the raw list? Refer to the shape, peaks, and spread of your display.	To organise the 40 prices, I first found the minimum value (KSh 82) and the maximum value (KSh 124), giving a total range of KSh 42. I chose five equal class intervals of width KSh 9 — namely 80–88, 89–97, 98–106, 107–115, and 116–124 — because this number of classes is manageable and each interval has a reasonable frequency. I then went through the raw list using a tally, counting how many prices fell into each class. The resulting frequency table showed that the interval 98–106 had the highest frequency (14 out of 40 prices), while the extreme intervals at either end had the fewest entries. When I drew the histogram — with price intervals on the horizontal axis and frequency on the vertical axis — the shape of the distribution became immediately visible. The bars formed a roughly bell-shaped (symmetric) mound centred around the 98–106 interval. This shape was completely invisible in the raw list. The histogram showed that most prices cluster tightly in the middle range (roughly KSh 98–KSh 115), with only a few prices at the very low or very high extremes. This visual 'aha' moment explains why the neighbour feels the price is always about the same — most of the time, it genuinely is near the centre — while also confirming the family's experience, because extreme prices do occasionally occur at the tails of the distribution.

Section 3 — Calculating and Interpreting Measures of Central Tendency and Spread

Calculate (or clearly describe) the mean, median, and mode of the 40 maize flour prices. Then calculate the range. For each measure, write one sentence explaining what it tells us about the prices at Wakulima Market. Which measure of central tendency gives the most honest and useful summary for a Nairobi family trying to budget	After organising the 40 prices, I calculated the following measures: Mean: I added all 40 prices and divided by 40. The mean was approximately KSh 104. This tells us that if the total cost of all 40 days of maize flour were shared equally across every day, each day would cost about KSh 104. Median: I arranged the 40 values in ascending order and found the average of the 20th and 21st values. The median was KSh 103. This tells us that exactly half of the recorded prices were below KSh 103 and half were above it — it is the 'middle' price. Mode: The most frequently occurring price in the raw data was KSh 100 (appearing 5 times). This tells us the single price that occurred most often at the stall.

for maize flour — and why? Are there situations where a different measure might be more appropriate?	Range: KSh 124 − KSh 82 = KSh 42. This tells us that prices varied by as much as KSh 42 from the cheapest to the most expensive day — a significant swing that would matter to a family on a tight budget. For a family budgeting for maize flour, the median (KSh 103) is the most honest and useful measure, because it is not distorted by the occasional extreme high or low price. If there were a few very high outlier prices, they would pull the mean upward and make the budget seem higher than what the family normally pays. The mode is less useful here because it only reflects the most common single price, not the overall pattern. However, for a market trader at Wakulima Market who wants to know their most common selling price in order to set a standard price tag, the mode would be the most appropriate measure.

Section 4 — Resolving the Mystery: How Can Both Be Right?	
Use your statistical evidence — your measures of central tendency, your range, and your data display — to write a clear explanation of why BOTH the Nairobi family AND their neighbour can be telling the truth at the same time. Your explanation should show that you understand the difference between the centre of a distribution and its spread. Then explain what an 'honest' use of statistics looks like — how could someone misuse these same numbers to tell a misleading story?	Both the family and the neighbour are telling the truth — they are simply describing different statistical features of the same data set. The neighbour is right about the CENTRE. Our data shows that the mean is KSh 104 and the median is KSh 103. The histogram reveals that the majority of the 40 prices — roughly 28 out of 40 — fall within a relatively narrow band of KSh 95 to KSh 115. If you shop at Wakulima Market on a typical day, you will most likely pay close to KSh 103–104. From this perspective, the price is 'always about the same.' The family is right about the SPREAD. Our range of KSh 42 shows that prices can swing from as low as KSh 82 to as high as KSh 124. If the family happened to shop on a day when prices were at KSh 82 and also on a day when prices reached KSh 124, they experienced a difference of KSh 42 — that genuinely feels like wild variation, especially if their weekly budget is tight. Both experiences are real and valid. The lesson here is that to tell an honest story about data, you must always report BOTH a measure of centre AND a measure of spread. Reporting only the mean (KSh 104) without mentioning the range (KSh 42) hides important information. This is how statistics can be misused: a market authority could report 'the average maize price is KSh 104' to make things sound stable, while concealing that prices spike to KSh 124 on certain days — a fact that matters enormously to low-income families. An honest statistician presents both the centre and the spread, clearly labels their graphs, states the size of their data set, and explains the limitations of their summary.

Section 5 — Using Statistics to Support a Real Decision	
Imagine you are presenting your findings to ONE of the following people: (a) a parent in Nairobi who needs to set a weekly budget for maize flour, (b) a stall owner at Wakulima Market who wants to set a fair and competitive price, or (c) a	EXAMPLE — addressed to a parent budgeting for maize flour: Dear Parent, Based on 40 days of recorded maize flour prices at Wakulima Market, I can help you plan your household budget more accurately. The median price for a 2 kg packet is KSh 103, meaning that on most shopping trips you should expect to pay close to this amount. However, you should be aware that prices can vary by as much as KSh 42 — dropping as low as KSh 82 on some days and rising

county government officer in Nairobi who monitors food prices to protect consumers. Choose ONE person. Write a short, clear statistical report (3–5 sentences) addressed directly to them, using your data to give them specific, actionable advice. Your report must include at least one number from your analysis and must connect statistics to a real-world decision.	as high as KSh 124 on others. I recommend that you budget KSh 115 per packet to comfortably cover most price variations, and if you find prices above KSh 115 on any given day, it may be worth waiting or trying another stall. Shopping earlier in the week, when our data shows slightly lower prices, could also save your family money over time. Yours faithfully, [Student Name], Grade 10 Statistics Analyst --- NOTE TO STUDENTS: Your report does not have to match this example exactly. What matters is that you use a real statistic from your work, address your chosen audience clearly, and give advice that is genuinely useful and grounded in the data.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Statistical Knowledge & Accuracy	Mean, median, mode, and range are all correctly calculated or described with working shown. All statistical terms are used accurately and precisely throughout. No errors in computation or definition.	At least three of the four measures are correctly calculated with minor errors in one. Statistical terms are mostly used correctly with one or two imprecise usages.	Only one or two measures are attempted or there are significant errors in calculation. Statistical vocabulary is limited or frequently misused.
Data Organisation & Graphical Display	Frequency table is correctly constructed with appropriate class intervals and accurate tallies. Histogram or frequency polygon is correctly drawn with labelled axes, consistent scale, and accurate bars/points. The graph is clearly connected to the analysis in the written explanation.	Frequency table is mostly correct with minor errors in one class interval or tally. Graph is drawn with labels but has minor scaling or accuracy issues. Some connection is made between the graph and the written analysis.	Frequency table has significant errors or is incomplete. Graph is missing, unlabelled, or does not accurately represent the frequency table. Little or no connection is made between the visual display and the analysis.
Conceptual Understanding of Centre vs. Spread	The explanation of why both the family and the neighbour can be right is clear, logically structured, and explicitly links the neighbour's view to a measure of central tendency and the family's view to the range/spread. The distinction between centre and spread is demonstrated with specific numerical evidence from the data.	The explanation acknowledges that both parties can be right and references centre and spread, but the connection to specific statistics is partially developed or not fully precise. The logic is mostly clear.	The student acknowledges a disagreement but cannot clearly explain how both can be right using statistical reasoning. Little or no reference is made to measures of centre versus spread.
Honest & Critical Use of Statistics	The student explicitly identifies at least one way statistics could be misused or presented misleadingly (e.g. reporting	The student shows awareness that statistics can be presented in different ways and makes some comment on	The student does not address the honest or critical use of statistics, or only makes a vague comment without connecting it to

	mean without range, cherry-picking data). The student articulates what makes a statistical report honest and applies this to the Wakulima Market context with specific examples.	honesty or misuse, but the example or explanation is general rather than specific to the data.	the data or context.
Real-World Application & Communication	The statistical report in Section 5 is clearly addressed to the chosen audience, uses an appropriate and respectful tone, includes at least one specific statistic, and gives actionable advice that a real person could act on. The writing throughout the document is organised, clear, and accessible to a non-specialist reader.	The report addresses the chosen audience and includes a statistic, but the advice is somewhat general or the tone is not fully appropriate for the audience. The overall writing is mostly clear with minor lapses in organisation.	The report does not clearly address a specific audience, omits relevant statistics, or gives advice that is too vague to be actionable. Writing is difficult to follow or poorly organised.